The earliest recorded female mathematician, Hypatia advanced the concept of conics, which underpins geometry and astronomy, and has important applications in everything from bridge building and architecture to satellite navigation.

MILESTONES

SCHOOL DIRECTOR
Becomes director of Alexandria's Neoplatonic School in c.400 CE, where she also teaches.

SEMINAL WORKS
Writes commentaries on Apollonius's *Conics* and Diophantus's *Arithmetica* in the early 400s CE.

SCIENTIFIC MARTYR
Perceived as a symbol of scientific paganism; dies at the hands of a conservative Christian mob in 415 CE.

TEXTS PRESERVED
Work survives to the 10th century and is recorded in the *Suda*, an encyclopedia of the Byzantine world.

ARTISTIC TRIBUTE
She is the only woman depicted in Raphael's 1511 Renaissance painting *School of Athens*.

Hypatia's mathematical genius may in part be explained by the outstanding education she is thought to have received from her father, Theon, a professor of mathematics at the University of Alexandria in Egypt, then part of the Roman Empire. She collaborated with her father on a treatise about Greek mathematician Euclid and probably helped him revise a groundbreaking 13-volume work, Euclid's *Elements*, which sets out extensive constructions and proofs for geometry, proportion, and number theory. Hypatia may also have assisted her father in preparing his edition of *The Almagest* by the ancient Greek mathematician Ptolemy. As well as teaching his daughter mathematics, astronomy, astrology, and philosophy, Theon also impressed upon her the significance of teaching, and she developed into a gifted speaker. This was an important factor in one of Hypatia's major achievements: the popularization of mathematics.

Math and mechanics

Although she also lectured on astronomy, philosophy, and mechanics, Hypatia became best known for her mathematics work—in particular, her explanation of the conic sections introduced by Apollonius in 200 BCE. Conic sections are one of the most important functions of geometry; they are produced when a plane intersects a cone. Depending on the angle, the resulting conic section may be a circle, ellipse, parabola, or hyperbola. Hypatia edited *On the Conics of Apollonius*, which elaborated on conic sections and made them easier to understand and apply. As a result, conics became firmly entrenched in the mathematical repertoire.

Hypatia may have developed *the astrolabe, an astronomical calculating device used to measure the distances of stars to establish the user's latitude.*